STROBE Statement—Checklist of Items for Observational, Functional Genomic, and Interventional Studies

Triangulating Single-Cell Perturbation Screens with Epigenetic Priors Reconstructs Causal Super-Enhancer Regulatory Networks in Triple-Negative Breast Cancer

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Section / Topic	Item #	STROBE Checklist Recommendation	Detailed Implementation in TNBC Causal ML Manuscript	Manuscript Location
Title and Abstract	1a	Indicate study's design with a commonly used term in title/abstract	Title and Abstract clearly indicate the study design: genome-scale single-cell CRISPRi perturbation screening, multi-omics epigenetic prior integration, causal DAG triangulation, and clinical survival validation in human TNBC cohorts.	Title; Abstract; Page 1
Title and Abstract	1b	Provide an informative and balanced summary of what was done and found	Structured abstract delineates clinical rationale (TNBC chemo-refractoriness), dataset scale (GSE224986; 95,678 cells, 11,975 sgRNAs), causal triangulation methodology, master driver prioritization (JUND, MYC, EP300, CTCF), TreeExplainer SHAP attribution, benchmarking against existing GRN methods, in silico KO validation ($R^2 = 0.473$), clinical survival (TCGA-BRCA, METABRIC), and epigenetic combination therapies.	Abstract; Page 1
Introduction - Background	2	Explain scientific background and rationale for investigation	Section 1.1–1.3: Aggressive clinical trajectory and lack of targeted receptor therapies in TNBC; high reliance on distal super-enhancers; mathematical and biological failure of observational transcriptomic GRNs (GENIE3, SCENIC) driven by dropout zero-inflation (>45% FDR); necessity of interventional do-calculus and functional single-cell CRISPRi.	Sections 1.1–1.3; Pages 1–2
Introduction - Objectives	3	State specific objectives, including any prespecified hypotheses	Section 1.4: Explicitly defines 6 primary investigative objectives: (i) construct tripartite causal GRN (TF → Enhancer → Gene); (ii) prioritize master drivers via PageRank and MRS; (iii) decode non-linear cooperativity via Explainable AI (SHAP); (iv) benchmark against state-of-the-art GRN algorithms; (v) validate clinical prognosis in independent patient cohorts; and (vi) map synergistic epigenetic combination therapies.	Section 1.4; Page 2
Methods - Study Design	4	Present key elements of study design early in the paper	Section 2.1–2.4: Comprehensive end-to-end causal machine learning workflow uniting physical chromatin occupancy (ENCODE Factorbook ChIP-seq) with randomized interventional perturbation response (single-cell CRISPRi / dCas9-KRAB).	Sections 2.1–2.4; Pages 2–3

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Methods - Setting	5	Describe setting, locations, relevant dates, exposure periods	Section 2.1: In vitro model system MDA-MB-231 (archetypal mesenchymal/basal-like TNBC); genome-scale single-cell CRISPRi Perturb-seq screen across 8 independent 10x Genomics Chromium 3' sequencing lanes (GEO accession GSE224986).	Section 2.1; Page 2
Methods - Participants / Cells	6a	Give eligibility criteria, sources, and selection methods	Section 2.1: Stringent single-cell quality control thresholds: cells retaining >= 200 detected genes and <= 15% mitochondrial reads; genes expressed in >= 3 cells; resulting in 95,678 high-quality single cells and 32,799 gene features.	Section 2.1; Table 1; Figure 1
Methods - Variables	7	Clearly define all outcomes, exposures, predictors, confounders	Sections 2.1–2.4: Exposure = targeted dCas9-KRAB single-guide enhancer silencing (819 somatic mutation loci); Outcome = log2 fold change and Welch t-statistic of target genes; Predictors = ChIP peak confidence score C_ChIP(i, e) in [0, 1000] and causal weight W_causal; Potential Confounders = technical dropout zeros, 10x sequencing lane batches, and multi-guide epistatic crosstalk.	Sections 2.1, 2.3, 2.4; Table 1
Methods - Data Sources	8	Give sources of data and details of measurement methods	Sections 2.1, 2.2, 2.7: GEO GSE224986 (single-cell CRISPRi), GSE223812 (enhancer coordinates), UCSC / ENCODE encRegTfbsClustered (ChIP-seq priors across 340 TFs), TCGA-BRCA (n = 160 TNBC) and METABRIC (n = 320 TNBC) clinical cohorts.	Sections 2.1, 2.2, 2.7; Pages 2–4
Methods - Bias Control	9	Describe any efforts to address potential sources of bias	Sections 2.1, 2.3, 2.4: Unconfounded causal identifiability enforced by evaluating single-guide perturbation cells (n = 3,930) strictly against unperturbed negative-control cells (n = 7,767), breaking multi-guide epistatic confounding; observational dropout zeros overcome by requiring physical ChIP-seq binding priors; lane batch effects controlled.	Sections 2.1, 2.4; Supplementary Note 2
Methods - Study Size	10	Explain how study size was arrived at	Section 2.1: Complete experimental library of 95,678 single cells, 11,975 unique sgRNAs, 819 targeted super-enhancers, and 32,799 gene features; stratified into single-guide (n = 3,930), multi-guide (n = 83,981), and control (n = 7,767) cohorts.	Section 2.1; Table 1; Figure 1d
Methods - Quantitative Variables	11	Explain how quantitative variables were handled in analyses	Sections 2.3–2.6: Library size normalization to 10,000 counts per cell (CP10k); log1p transformation; vectorized Welch t-statistic computation with Welch-Satterthwaite degrees of freedom; composite causal weight scaling W_causal = C_ChIP × |log2 FC| × |t|.	Sections 2.3, 2.4; Supplementary Notes 1, 3
Methods - Statistical Methods	12a	Describe all statistical methods, including confounder control	Sections 2.3–2.7: Fast vectorized two-sample Welch t-test; Google PageRank graph centrality (damping factor $\alpha = 0.85$); Gradient Boosted Decision Trees (XGBoost Regressor; 100 estimators, depth = 4); exact TreeExplainer SHAP attribution; multivariable Cox proportional hazards modeling adjusted for age, tumor stage, and nodal status.	Sections 2.3–2.7; Supplementary Notes 3–5
Methods - Subgroups &	12b	Describe methods examining subgroups	Sections 2.3, 2.8: Unsupervised Leiden community detection (resolution = 0.5)	Sections 2.3, 2.8;

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Interactions		and interactions	resolving 12 distinct transcriptional phenotypic states (Clusters 0 to 11); multi-factor combinatorial epigenetic perturbation simulations (BETi + HATi + HDACi).	Figures 1c, 8
Methods - Missing Data Handling	12c	Explain how missing data were addressed	Sections 2.1, 2.4: Zero-inflation handled naturally via interventional differential testing without artificial expression imputation; chunked Compressed Sparse Row (CSR) matrix representation in h5py and AnnData.	Sections 2.1, 2.4; Pages 2–3
Results - Participants / Cells	13a	Report numbers of individuals/cells at each stage of study	Section 3.1: 95,678 single cells evaluated post-QC; 32,799 expressing genes; stratified across 12 Leiden clusters and 8 experimental sequencing lanes; 3,930 single-guide cells, 83,981 multi-guide cells, 7,767 negative-control cells.	Section 3.1; Table 1; Figure 1
Results - Descriptive Data	14a	Give characteristics of study participants/cells	Section 3.1, Table 1, Figure 1: Median 6,519 UMIs per cell (Figure 1a); median 2,138 detected genes per cell (Figure 1b); 819 targeted super-enhancer loci across 11,975 sgRNAs.	Section 3.1; Table 1; Figure 1a–d
Results - Main Findings	16a	Give unadjusted and adjusted effect estimates and precision	Sections 3.2–3.5, Tables 2–5, Figures 2–5: 15,059 physical TF–enhancer prior edges (Table 2); 1,110,566 functional perturbation responses (Section 3.3); 6,366,350 tripartite causal paths (Section 3.4); master drivers CTCF (118.7k paths), POLR2A (110.5k), JUND (75.3k), EP300 (67.3k), MYC (54.8k); TreeExplainer SHAP attribution ranking JUND (mean |SHAP| = 0.0296) and MYC (0.0174) with non-linear switch dynamics (Table 5).	Sections 3.2–3.6; Tables 2–5; Figures 2–6
Results - Benchmarking & Validation	17	Report other analyses done (sensitivity, benchmarking, validation)	Sections 3.7–3.8, Tables 6–7, Figure 7: Causal Triangulation AUROC = 0.941 and AUPRC = 0.918 vs SCENIC (0.782 / 0.694), CellOracle (0.751 / 0.648), Scribe (0.688 / 0.582), PIDC (0.642 / 0.521); in silico driver knockout validation yielding Pearson r = 0.688, R^2 = 0.473 ($p = 3.37 \times 10^{-122}$, N = 864 genes); clinical validation in TCGA-BRCA (n = 160) and METABRIC (n = 320) basal-like cohorts.	Sections 3.7–3.8; Table 6; Figure 7a–e
Discussion - Key Results	18	Summarize key results with reference to study objectives	Section 4.1–4.3: Successfully triangulated 6.36M directed causal edges; eliminated observational confounding and high false-positive rates; established continuous DAG identifiability; unmasked non-linear AP-1/MYC master regulatory axes; identified tri-modal epigenetic synergy.	Sections 4.1–4.3; Pages 6–7
Discussion - Limitations	19	Discuss limitations of study (potential bias, imprecision)	Section 4.4: In vitro cell-line setting (MDA-MB-231); targeted silencing of non-coding enhancers rather than direct genetic coding knockouts of upstream transcription factor genes; computational rather than in vivo clinical trial validation of predicted combination therapies.	Section 4.4; Page 7
Discussion - Interpretation	20	Give a cautious overall interpretation of results	Sections 4.2–4.3: Triangulating randomized Perturb-seq interventions with physical ChIP occupancy provides a mathematically defensible, biologically validated approach for cancer gene regulatory network discovery and overcoming drug resistance in TNBC.	Sections 4.2–4.3; Page 7

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Funding & Disclosures	22	Give source of funding and role of funders	Section 5 (Declarations): Supported by National Science & Technology (NST) Fellowship (2025–2026), Ministry of Science and Technology, Government of the People's Republic of Bangladesh (Awardee: Omar Faroque, Group: Physical & Mathematical Sciences). Funders had no role in study design, data collection, analysis, or manuscript preparation.	Declarations; Page 7

Note: This checklist adheres to the STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) guidelines adapted for molecular epidemiology, high-throughput single-cell functional genomics, and interventional causal discovery. All metrics, cohort dimensions, and experimental parameters strictly match the primary manuscript text, figures, and tables.